Flow Control at Data Link Layer using Sliding window protocol

EXPT

Write

AIM:

To write a program to implement flow control at the Data Link Layer using the sliding window protocol and to simulate the flow of frames from one node to another.

ALGORITHM:

* Start the program
* Read the total number of frames to be sent (N), the sender's window size (w), and (optionally) a frame number to simulate as lost during transmission.
* Initialize base = 0 (sequence number of the oldest unacknowledge frame) and nextseq = 0.
* Repeat while base < N:
  i) Determine the current sender window as the range [base, base + W - 1].
  ii) Send all frame with sequence number from nextseq up to (base + w - 1).
  iii) If the lost frame lines within the frames
* Else (no loss in this batch):
  i) The receiver accepts each frame in order and sends a cumulative acknowledgement (ACK) for it.
  ii) On receiving the ACK for the last frame sent, the sender slides its window forward.
  base = nextseq

== SLIDING WINDOW PROTOCOL SIMULATION (Go-Back-N) ==

Enter total number of frames to send: 8

Enter sender's window size: 4

Enter frame number to simulate as LOST

(0 to 7) enter -1 for no loss): 5

Sender window: [Frame 0 – Frame 3]

Node A --> Sending Frame 0

Node A --> Sending Frame 1

Node A --> Sending Frame 2

Node A --> Sending Frame 3

Node B --> Received Frame 0, Sending ACK 0

Node B --> Received Frame 1, Sending ACK 1

Node B --> Received Frame 2, Sending ACK 2

Node B --> Received Frame 3, Sending ACK 3

Node A --> ACK 3 received. Sliding window forward.

Sender window: [Frame 4 – Frame 7]

Node A --> Sending Frame 4

Node A --> Sending Frame 5

(Frame 5 LOST in transit)

Node A --> Sending Frame 6

Node A --> Sending Frame 7

Node B --> Frame 5 missing. Discarding out-of-order frames after.

*Repeat until N frames have been transmitted and an acknowledged.

* Stop the program.

PROGRAM:-

```
def simulate(total_frames, window_size, lost_frame):
def simulate(n, w, lost):
    base = nxt = 0
    while base < n:
        print("\n----------")
        print(f"Sender Window: [Frame {base} - Frame
                {min(base+w-1, n-1)}]")
        while nxt < min(base+w, n):
            print(f"Node A -> Sending Frame {nxt}")
            if nxt == lost:
                print(f"(Frame {nxt} LOST in transit!)")
            nxt += 1
        if lost is not None and base <= lost < nxt:
            print(f"Node B --> Frame {lost} missing. Discarding
                    out of order frames after")
            print(f"Node A --> Timeout! Resending window
                    starting from Frame {lost}.")
            nxt, lost = lost, None
            continue
        for i in range(base, nxt):
            print(f"Node B --> Received Frame {i},
            print(f"Node B --> Received Frame {i},
```

Sender window : [Frame 4 - Frame 7]

Node A --> Sending Frame 5

Node A --> Sending Frame 6

Node A --> Sending frame 7

Node B --> Received Frame 4, Sending ACK 4

Node B --> Received Frame 5, Sending ACK 5

Node B --> Received Frame 6, Sending ACK 6

Node B --> Received Frame 7, Sending ACK 7

Node A --> ACK 4 received. Sliding window forward.

All 8 frames successfully transmitted from Node A to Node B.

```
        base = nxt
    print("\n ----------")
    print(f"All {n} frames successfully transmitted from Node
                A to Node B.")

print("== SLIDING WINDOW PROTOCOL - FLOW CONTROL
                SIMULATION (Go-Back-N) ==\n")
n = int(input("Enter total number of frames to send: "))
w = int(input("Enter sender's window size: "))
lost = int(input(f"Enter frame number to simulate as
            LOST \n (0 to {n-1}, enter -1 for no loss): "))
simulate(n, w, None if lost == -1 else lost)
```

(80)

RESULT:

The program to implement flow control at the Data Link layer using the sliding window Protocol was successfully written in Python, executed and verified by simulating the flow of frames between two nodes (Node A as sender and Node B as receiver).